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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: PROJECT EVALUATION (24CS2235F) A.Y. 2026 - 2027

TITLE OF THE PROJECT	Parse Tree Generator GUI	
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Abstract

Compiler Design is one of the fundamental subjects in computer science, where understanding grammar analysis, parsing techniques, and syntax tree construction plays a vital role in developing programming language compilers. However, performing grammar transformations and parsing operations manually is often complex, time-consuming, and prone to errors, making it difficult for students to understand these concepts effectively. Existing parser tools mainly focus on parser generation and provide limited support for integrated grammar analysis and interactive visualization.

This project presents a **Grammar Analyzer and Parse Tree Generator using LL(1) Parsing**, developed using **Python** and **Tkinter**. The application accepts Context-Free Grammar (CFG) rules and an input string as input. It performs grammar validation, detects and removes direct and indirect left recursion, applies left factoring, computes FIRST and FOLLOW sets, generates the LL(1) parsing table, validates the input string, and produces leftmost and rightmost derivations. Finally, the system constructs and graphically visualizes the parse tree using color-coded nodes to distinguish terminals and non-terminals, making compiler concepts easier to understand.

The proposed system provides an interactive and user-friendly learning environment that automates major compiler design tasks while reducing manual effort. By combining grammar analysis, parsing, derivation generation, and graphical visualization in a single application, the project enhances the learning experience for students and serves as an effective educational tool for Compiler Design courses

Signature of the Guide HOD